

ANUJ KUMAR

Embedded Systems & Software Engineer

+91-9044292097 | anujyadav11112003@gmail.com | [GitHub](#) | [LeetCode](#) | [LinkedIn](#)

SUMMARY

Embedded Systems Engineer specializing in real-time system optimization and low-level software architecture. Proficient in C/C++, Linux kernel development, TCP/IP protocol stacks, and bare-metal firmware programming. Seeking to apply memory management and device driver expertise to develop high-performance mobile chipsets at Qualcomm.

TECHNICAL SKILLS

Languages & Core CS: C/C++, Embedded C, Data Structures & Algorithms, Object-Oriented Design (OOD), Python, SQL
Embedded & OS: Linux Kernel Development, Device Drivers, ARM Cortex-M, RTOS, IPC, 8051 MCU, UART, I2C, SPI
Networking & Systems: Communication Protocol Stack, TCP/IP, UDP, WebSockets, SIP/RTP, Multithreading, Memory Management
Tools & Frameworks: GDB, Valgrind, Git, Docker, Node.js, React, Redis, MongoDB, AWS, CI/CD pipelines

EDUCATION

JSS Academy of Technical Education

Noida, India

B.Tech in Electronics & Communication Engineering | Class XII (CBSE): 94.6%

Nov 2022 – Jul 2026

- **Relevant Coursework:** Embedded Systems, Microprocessors, Operating Systems (OS), Computer Networks, DSP, System Design

PROJECTS

ARM Cortex-M3 UART-I2C Protocol Bridge | *Embedded C, ARM CMSIS*

[Source Code](#)

- Developed real-time embedded software to translate UART to I2C, implementing interrupt-driven Rx lines with a lock-free circular buffer.
- Offloaded bulk data transfers to the DMA controller to accelerate peripheral I/O transactions, reducing ISR latency by **70%**.

Concurrent IPC Collaboration Engine | *C++, Node.js, WebSockets*

[Live](#)

- Engineered a high-concurrency server handling **50+ parallel TCP/WebSocket sessions** at sub-100ms transmission latency.
- Designed a decentralized delta-diff synchronization protocol to maintain data consistency and eliminate centralized thread-locking bottlenecks.

Vynkify – Network Routing Service | *Node.js, Redis, TCP/IP, MongoDB*

[Live](#)

- Built a resilient communication protocol stack handling 50K+ monthly traffic redirections through enhanced TCP/IP routing.
- Decreased lookup latencies by **60%** using targeted Redis memory caching and structural indexing.

LearnMate – Scalable Processing Platform | *Python, React, MongoDB*

[Live](#)

- Constructed a high-availability processing platform for **100+ active users**, implementing stateless session authentication with JWT tokens.
- Automated remote deployments using robust CI/CD pipelines, maintaining a verified production uptime of **99.9%**.

EXPERIENCE

Software Development Intern

Jan 2024 – Mar 2024

Quizzard

Remote

- Conducted runtime profiling to eliminate execution bottlenecks, reducing system latency by **35%** for 10K+ active users.
- Architected a modular software framework using Object-Oriented Design (OOD) principles to accelerate team deployment velocity by **40%**.
- Streamlined network efficiency using request batching and data caching to decrease API communication overhead by **30%**.

CERTIFICATIONS & ACHIEVEMENTS

Data Structures & Algorithms: Engineered optimized solutions for **1000+ complex algorithmic challenges** on LeetCode and CodeChef, prioritizing minimal time complexity and efficient memory allocation.

Certifications: **7x AWS Certified** (including DevOps Professional, ML Specialty) | **5x FreeCodeCamp Certified**.